

# The Complete Project

AI-Based Automatic Block Planning · Chennai Division  
Narrative, diagrams, illustrations, and the honest roadmap

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**Purpose** The single document that explains everything — in words, in diagrams, and in pictures

**Companion documents** 1. System Blueprint (engineering) · 2. PRD & Data Specification

**Includes** What was deliberately simplified, what was consciously excluded, and what comes next

**Version** 2.0

**Read this one first.** Documents 1 and 2 are reference material for people building the system. This document is for anyone who wants to understand it — the problem, the reasoning behind each design decision, what the interfaces look like, how it is evaluated, how it is demonstrated, and where its limits genuinely lie.

## Contents

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- 1 The problem, in plain language
- 2 The coordination gap — the one image that matters
- 3 Why this became urgent
- 4 How the system works — the narrative walkthrough
- 5 Representing the railway — the network model
- 6 Finding windows and finding merges
- 7 Deciding — the cost of everything in one unit
- 8 Where the intelligence actually lives
- 9 Planning across horizons
- 10 When reality breaks the plan
- 11 The scenario, end to end
- 12 The five points of view, illustrated
- 13 Ethics and responsible design
- 14 Proving it works
- 15 The demonstration narrative
- 16 **Deliberate simplifications, exclusions and future extensions**
- 17 Honest limitations
- 18 Glossary

## 1 The problem, in plain language

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Three separate departments look after the parts of a railway that do not move. **Engineering** keeps the track: rails, sleepers, ballast, bridges, points and crossings. **Traction Distribution** keeps the overhead electrical equipment: masts, contact wire, insulators, substations. **Signal & Telecommunication** keeps the signalling: signals, point machines, track circuits, axle counters, interlocking.

To work on any of it, the line usually has to be taken out of service for a while. There are three ways this happens, and the distinction matters:

| Mechanism            | Who needs it          | What it means                                                                                                                            |
|----------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Traffic block</b> | Engineering           | No trains on that line for a defined period                                                                                              |
| <b>Power block</b>   | Traction Distribution | Overhead wire de-energised. On an electrified line this quietly implies a traffic block too — electric trains cannot run under dead wire |
| <b>Disconnection</b> | Signal & Telecom      | Signalling gear temporarily disconnected from the interlocking, with the Station Master's authority                                      |

Each department keeps its own list of what needs doing — defects found on inspection, maintenance that has fallen overdue — in its own system. Engineering has TMS, S&T has SMMS, Traction Distribution has TDMS. When a department wants track access, it files a request in BDMS. The Control Office, through COA, knows when the corridor is actually free.

**And that is the whole problem.** Those three request streams never meet. Nobody merges them. There is no screen anywhere on which an Engineering officer can see that Traction Distribution wants the same four kilometres of line next Thursday.

**One thing this is not about.** This concerns **fixed infrastructure** — things bolted to the ground. It is not about coaches. Coach maintenance, periodic overhauls and shed scheduling belong to the Mechanical department, run on entirely different systems, and need no block at all: the coach comes to the shed, and the shed does not stop trains from running. The five systems named in the problem — TMS, SMMS, TDMS, COA, BDMS — do not touch coach maintenance.

## 2 The coordination gap — the one image that matters

Take three real jobs that happen to sit within four kilometres of one another on the same stretch of line: a three-hour Engineering job, a ninety-minute Traction Distribution job, and a one-hour S&T job. Planned independently they take three separate windows on three separate nights.

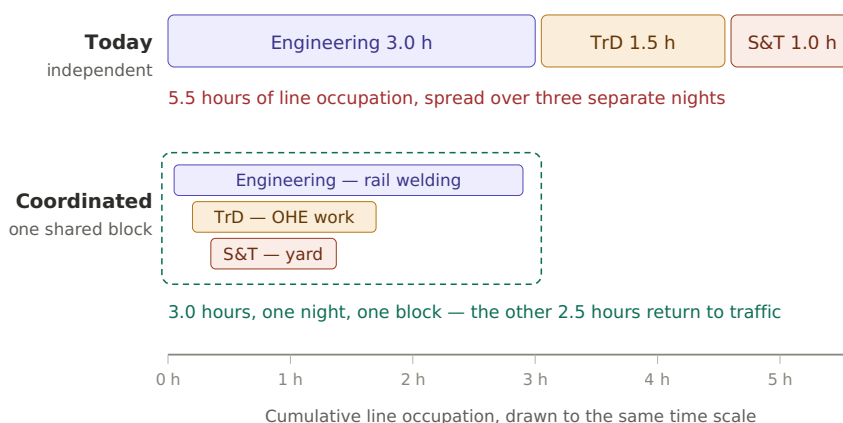


Figure 1 — The same three jobs, planned independently and planned together.

The upper row lays them end to end to show **cumulative line occupation** — the total time the corridor is unavailable. Because each department files its own request, the corridor absorbs the sum: five and a half hours.

The lower row shows them inside one window. Engineering occupies almost the full three hours. Traction Distribution's overhead work runs alongside it on the same de-energised stretch. S&T's yard disconnection runs concurrently at the station. The block ends when the longest job ends. **Two and a half hours returned to train operations, with no maintenance sacrificed.**

**Everything else in this project exists to find and execute these merges reliably, at scale, without ever compromising safety.**

## 2.1 Why it is harder than "group things by location"

| Complication                              | What it means                                                                                                                                                             |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Priorities differ                         | A safety-critical rail flaw and a routine inspection do not deserve equal treatment                                                                                       |
| Some pairs must never share               | A point machine cannot be disconnected while heavy track machines work over that same turnout. The system needs rules about what <i>cannot</i> combine, not only what can |
| Traffic density varies                    | The same block costs twenty times more at 10:00 than at 02:00                                                                                                             |
| Line configuration changes everything     | On a quadruple-line section, blocking one line leaves three. On a single-line section, blocking it stops everything                                                       |
| Resources are scarce                      | A window may be free and a crew ready, but if the tamping machine is elsewhere the work cannot happen                                                                     |
| Requested duration is not actual duration | Work requested for four hours may historically take 4.7. And a granted block may be wasted entirely because nobody was ready                                              |
| Emergencies interrupt                     | A rail fracture consumes a window without warning                                                                                                                         |
| Seasons impose limits                     | Certain track work is not permitted during monsoon                                                                                                                        |

## 3 Why this became urgent

The uncoordinated habit is not new, and it used to be harmless. Sections carried far less traffic. There was slack in the day. A department could get a block more or less whenever it wanted, and nobody needed to coordinate because the corridor was rarely contested.

That has changed. Train density has risen, freight has grown, timetables have tightened, and free corridor windows have shrunk. **Block availability has itself become the scarce resource.**

**The framing that lands.** The argument is not “departments do not talk to each other.” It is: *the scarcity changed, and the planning process did not.* That is accurate, respectful of the people currently doing this work, and identifies precisely what the system fixes.

### 3.1 Why Chennai Division

Roughly 698 route kilometres, 1,935 track kilometres, 160 stations, almost entirely electrified. Compact enough to model honestly — and containing every variant of the problem in one division:

| Corridor                               | Character                         | Why it earns its place                                                                                              |
|----------------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Chennai Beach – Tambaram               | Quadruple, saturated suburban     | Capacity utilisation near 69%. Almost no daytime windows — everything compresses into 01:00–04:00. The hardest case |
| Chennai Central – Arakkonam            | Quadruple trunk, mixed            | Four tracks to Arakkonam, two towards Katpadi. Rich in merge opportunities and machine-heavy work                   |
| Arakkonam – Kanchipuram – Chengalpattu | Single line, partly unelectrified | A block here halts movement completely. Proves block cost is not uniform                                            |
| Basin Bridge – Gummidipoondi           | Freight-dominant                  | Ennore and Chennai Port traffic. Exercises goods forecasting and power blocks                                       |

Two further local characteristics strengthen it: the northeast monsoon from October to December gives a genuine, locally credible seasonal constraint, and the mix of suburban EMU, long-distance express and port-bound freight on overlapping corridors makes the traffic model instructively non-trivial.

## 4 How the system works — the narrative walkthrough

At the highest level the system does five things: it pools everyone's pending work into one place, scores it, works out when the corridor is actually free, packs compatible jobs into shared blocks, and hands the result to the people who already hold the authority to approve it.

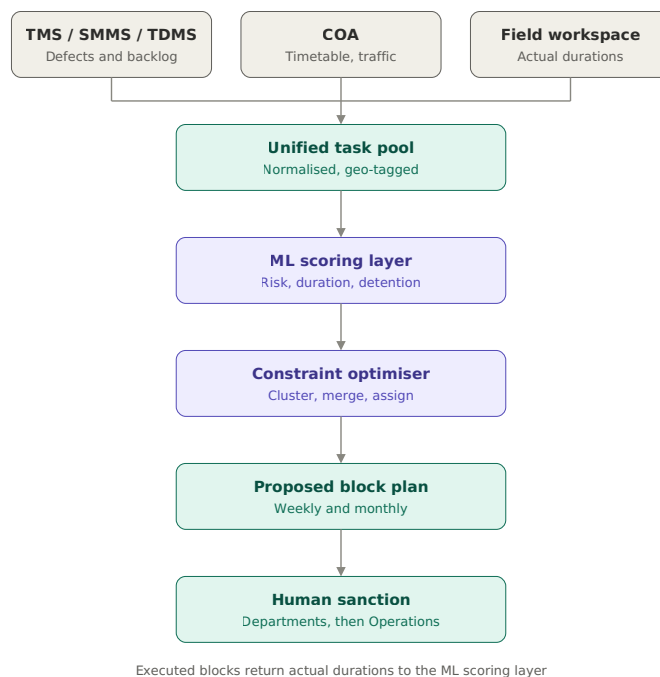


Figure 2 — The conceptual flow, top to bottom.

Three sources feed it. The departmental systems say what needs doing. COA says when the line is free. And the field application — the third input — says what actually happened last time.

**Why the field application is not a side feature.** No system in service today records how long a job *actually* took against how long it was *requested* for. Nobody logs “block granted 10:00–14:00, work started 10:40 because the tamper was late, finished 14:20.” That data does not exist — which is exactly why nobody can predict block durations today. The field app is the **data acquisition layer** that makes the learning possible at all.

## 4.1 The engineering shape of it

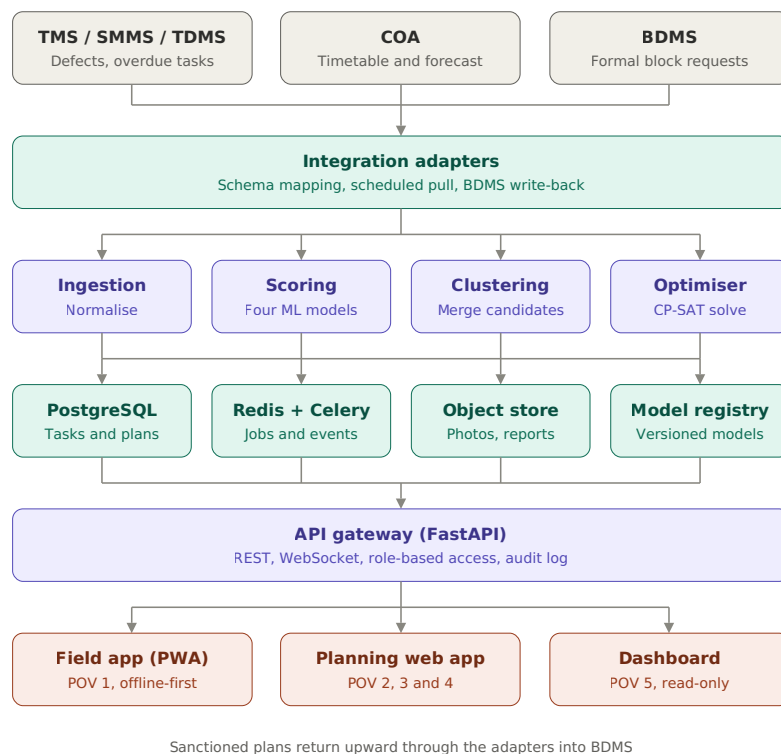


Figure 3 — The deployable component architecture.

Two things worth noticing. First, there are only **three client applications for five points of view** — the departmental officer, operations and coordination views are one web application under different roles. Building five separate frontends would triple the work for nothing.

Second, **BDMS appears at the top but the arrow at the bottom runs the other way**. The system reads pending requests from BDMS and writes sanctioned plans back into it. It is a planning layer *above* BDMS, not a replacement. On any conflict, BDMS wins. That single design decision is the difference between something Railways could deploy and something that would require Railways to abandon a working system.

## 4.2 The eight stages

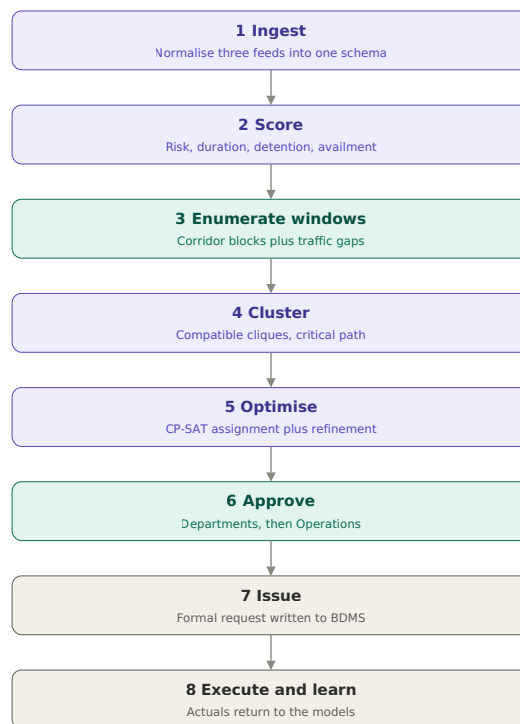
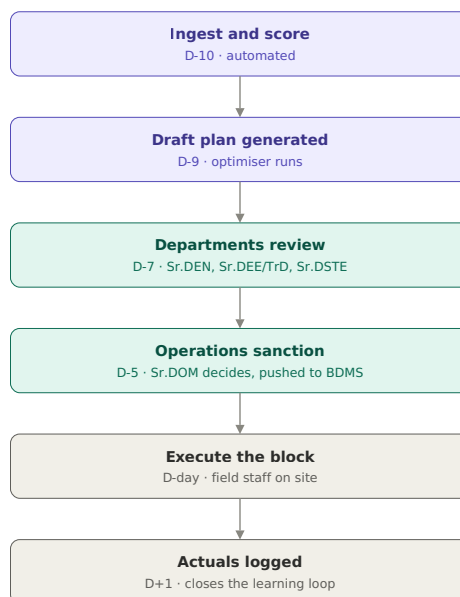


Figure 4 — Every feature belongs to one of these eight stages.

If a proposed feature does not fit somewhere on this spine, it should be cut.

## 4.3 The weekly rhythm



Emergencies interrupt and trigger re-planning at any stage

Figure 5 — The weekly planning cycle, from automated ingestion to the learning loop.

## 4.4 And the scale question, answered

Indian Railways is enormous as an institution, and the instinct is to assume the computation is correspondingly enormous. It is not. **The scale here is organisational, not computational.**

Block planning is never done at national level by anyone. It is done at **divisional** level — every officer who requests or sanctions a block holds a divisional post.



Figure 6 — How the problem size collapses.

A division carries roughly four hundred pending tasks in a week against a few hundred candidate windows. Naively that is a hundred thousand pairings — but **geographic locality** prunes it drastically. A rail flaw at km 42 between Tiruvallur and Arakkonam can only be scheduled into a window on that block section. In practice each task has five to fifteen feasible windows, leaving a few thousand decision variables. A constraint solver disposes of that in seconds.

**The data is not big either.** A division raises a few thousand tasks a year; five years of block history is tens of thousands of rows. Everything fits comfortably in memory. Putting distributed streaming infrastructure in the architecture would be conspicuous overkill, and a railway officer on a panel would recognise it as such.



## 5 Representing the railway — the network model

How you represent the geography determines whether merging works at all. The answer has three levels: **stations** are nodes, **block sections** are edges between consecutive block stations, and **corridors** are named runs of block sections used only for monthly planning and human conversation.

Tasks live on block sections, because that is the unit a block is actually granted over. A block is not sanctioned for “km 42.1” — it is sanctioned *between two stations, on a named line*. So merging needs no distance calculation: same block section, same line, one block covers both, whether the jobs are two hundred metres or eight kilometres apart.

### 5.1 It is a multigraph

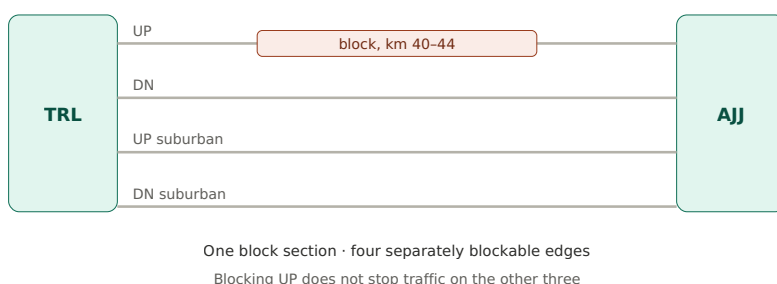


Figure 7 — Four parallel edges between two stations. A block takes one.

Between Tiruvallur and Arakkonam there are not one edge but four — up, down, and two suburban lines. Same endpoints, same kilometres, separately blockable. Blocking UP at 02:00 costs almost nothing because three edges remain. On a single-line section there is one edge, and blocking it severs the graph — nothing gets through. Same block, wildly different cost, and the graph is what knows the difference.

### 5.2 Two kinds of adjacency

- **Longitudinal** — the next section along the route. A train held here piles up there. This gives cascade delay.
- **Lateral** — the parallel edges within the same section. This gives rerouting: when UP is blocked, can trains work bidirectionally on DN?

They are kept separate. Conflating them causes either missed reroute options or double-counted delay.

### 5.3 Junctions, crossings and station limits

A fair objection: real track has junctions, crossovers, conflict zones and interlocking. Can parallel edges between stations really be enough?

The resolution mirrors the actual working rules: **tasks live either on an edge or on a node.**

| Where the task is      | How access is taken    | How it is modelled                                                                                                          |
|------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Mid-section plain line | Traffic or power block | Edge occupancy. This is the bulk of block-requiring work — tamping, screening, rail renewal, OHE inspection                 |
| Station or yard        | Disconnection          | A node resource model: the station has a route count, a disconnection consumes some, compatibility rules govern concurrency |

A junction such as Arakkonam is simply a node of degree greater than two. Block one arm and the others remain traversable — which is correct.

**The argument that settles it.** A block request is itself a station-to-station document. The model's resolution matches the artefact it produces. Modelling finer would need interlocking and route-table data that exists in none of the source systems, and would answer a question — minute-by-minute train regulation — that belongs to Control on the day, not to a weekly planner.

## 5.4 Two refinements that recover most of the realism

**Adjacent-line caution.** When men and machines work on UP, the parallel lines get a speed restriction. Charged over the **worksite length**, not the whole section — about a kilometre at 30 km/h instead of 110 gives roughly 1.4 minutes per train. Small individually, real when multiplied. Almost no student team models this, and a railway evaluator notices immediately.

**Single-line working is not free.** Blocking UP on a double-line section lets trains work bidirectionally on DN, but capacity drops sharply and crossovers are needed at both ends. Modelled as a capacity factor, not as costless rerouting.

## 6 Finding windows and finding merges

### 6.1 A gap is not a window

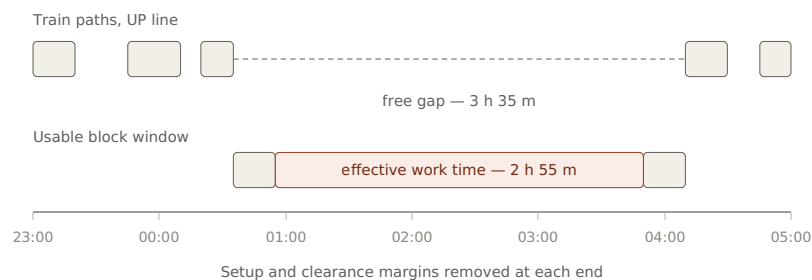


Figure 8 — Setup and clearance reduce a raw gap to usable work time.

The block does not begin the instant the last train clears. Men and machines have to get on track. At the other end the line has to be cleared, tools accounted for, and the section certified fit before traffic resumes. Both eat real time, which is why a twenty-five minute gap is worthless.

Windows come from two places: **pre-planned corridor blocks** already negotiated into the working timetable, and **traffic gaps** detected by scanning train paths. Three complications are modelled: goods trains are not in the fixed timetable and can fill an apparent gap; both lines of a double-line section cannot be blocked simultaneously without severing it; and gap availability varies enormously by line, since suburban lines only open up after midnight while trunk lines may have mid-morning holes.

**Corridor blocks come from a documented rule.** The real pattern is negotiated annually into the working timetable and is not available for this build, so it is **generated from a stated rule set held in configuration** — two hours per line per week at night on the trunk corridor, three hours twice weekly at night on suburban, three hours mid-morning weekly on the single line, two hours twice weekly mid-morning on the freight line. On deployment this is replaced wholesale by the division's actual corridor block table. No claim is made that these are Chennai's real corridor blocks.

### 6.2 Merging is clique-finding, and duration is a critical path

A valid merge needs **every pair** in the set to be compatible, not just each with the first. That is maximal clique enumeration — theoretically hard, practically instant at one block section with a handful of pending tasks. Candidates are capped at five, which is realistic; no genuine block carries ten departments.

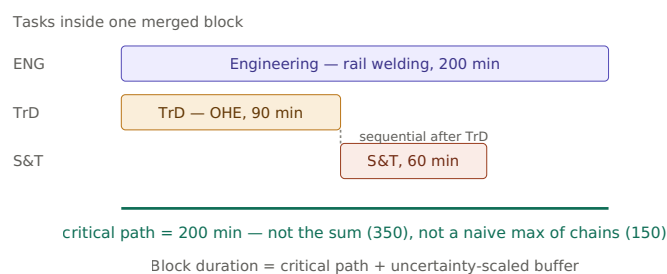


Figure 9 — Merged duration is the longest path through the precedence graph.

The naive answers are both wrong. A maximum gives 200 minutes and ignores the chain. A sum gives 350 and ignores the parallelism. The truth is that Traction Distribution feeds into S&T as a 150-minute chain running alongside Engineering's 200 — so the block is 200 minutes, plus buffer. Twenty lines of code, and it is the difference between a plan that holds and one that overruns every time.

### 6.3 Merging escalates what the block needs

| Tasks in the group                  | The merged block requires                                      |
|-------------------------------------|----------------------------------------------------------------|
| traffic + traffic                   | traffic block                                                  |
| traffic + power                     | traffic <b>and</b> power block                                 |
| traffic + disconnection             | traffic block, plus Station Master authority for the node work |
| power alone, on an electrified line | power block <b>implies</b> a traffic block anyway              |

Lines behave the same way. Three tasks all on UP is cheap. Two on UP and one on DN takes both lines, and on a double-line section that severs it. The cost model has to see that.

## 7 Deciding — the cost of everything in one unit

This is where “maximise asset availability” stops being a slogan and becomes arithmetic. Both sides of the trade-off reduce to the same thing:

**Cost of taking a block** = trains delayed now.

**Cost of not doing the work** = trains delayed later, when the defect forces a speed restriction.

Minimise the sum. Railway officers understand this immediately, because it is the currency they already think in.

### 7.1 What a block costs, worked through

|                               | Night: Wed 01:30–04:30   | Day: Wed 10:00–13:00             |
|-------------------------------|--------------------------|----------------------------------|
| Trains on UP                  | 2 goods, both divertible | 12; DN saturated, 5 must be held |
| Held cost                     | 0                        | ≈ 280                            |
| Reroute cost                  | ≈ 10                     | ≈ 78                             |
| Caution on parallel lines     | ≈ 8                      | ≈ 42                             |
| <b>Total weighted minutes</b> | <b>≈ 18</b>              | <b>≈ 400</b>                     |

Twenty-two times cheaper at night. The optimiser does not need to be clever to find that — but the cost function has to be able to see it.

### 7.2 Three terms that shape behaviour

| Term                        | What it makes the system do                                                                                                                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Window waste</b>         | Penalises unused window capacity. This is what actively <i>pushes</i> the solver toward merging rather than merely permitting it                                                              |
| <b>New access requested</b> | A soft penalty per newly requested window. Corridor blocks get used first; fresh access is asked for only when it genuinely pays. The policy lives in the cost function, not in a conditional |
| <b>Night-shift fairness</b> | Penalises uneven night work across crews. Cheap to add, and the difference between a system staff accept and one they resent                                                                  |

### 7.3 What a minute of delay is worth

Delay is weighted by train type. Two defensible positions:

| Train type                    | Aggressive punctuality | Balanced (chosen) |
|-------------------------------|------------------------|-------------------|
| Rajdhani class / Vande Bharat | 3.0                    | 1.8               |
| Superfast / Express           | 2.0                    | 1.5               |
| Suburban EMU                  | 1.5                    | 1.3               |
| Passenger                     | 1.0                    | 1.0               |
| Goods                         | 0.3                    | 0.7               |

The aggressive column mirrors how punctuality is officially measured. But pushed too far, freight becomes the dumping ground — every block lands on goods paths, port traffic to Ennore suffers, and you have optimised a metric rather than the railway. The balanced column is the default, and **the weights are visible in the interface** so officers can see what the system is valuing rather than having it buried.

## 7.4 Two rules that must never bend

**Marginal, not cumulative.** Deferring a task by one week adds only *that week's* incremental risk. Using cumulative escalation probability massively overstates it and makes the optimiser panic-schedule everything.

**Safety-critical work is a constraint, never a cost.** It carries a hard deadline and does not participate in the trade-off at all. A probability must never be allowed to weigh against a statutory limit. If the deadline cannot be met, the system escalates visibly rather than quietly rescheduling.

## 8 Where the intelligence actually lives

Four models, each solving something a lookup table cannot.

| Model           | What it predicts                                                  | Why it matters                                                                                                      |
|-----------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Duration        | Actual work minutes, as a distribution                            | Stops the plan collapsing on day one through systematic underestimation                                             |
| Escalation risk | Daily hazard that a defect becomes a failure or speed restriction | Turns “high / medium / low” into a defensible number, and supplies the marginal probability the cost function needs |
| Detention       | Weighted train delay a block will cause                           | Gives the optimiser a cost in a real unit                                                                           |
| Availment risk  | Probability the granted block is wasted                           | Modelled by no system in service today                                                                              |

### 8.1 Three details that make these credible

**Duration is predicted per task, not per block** — the merged duration is composed by critical-path logic. This also roughly triples the training data, since every task contributes a row whether it ran alone or merged. And one feature is essential: **whether the task was part of a merged block**. Merged work is often slower per task — more people on a confined worksite, shared access, coordination overhead. Omit it and the model systematically underestimates exactly the blocks the system exists to create.

**Escalation is survival analysis, not classification.** A defect attended on day six did not escalate — but it also never got the chance to. Treating that as a clean negative teaches the model that such defects are safe when they were actually rescued. So each defect is expanded into one row per day it stayed open, and the model learns a **daily hazard rate**. Compose hazards for any horizon, and the marginal probability falls out by construction.

**The subtle trap.** Dangerous-looking defects get fixed fastest, so in the raw data severe defects rarely escalate. Train naively and the model learns that severity 1 is safe — exactly backwards. Modelling hazard *conditional on remaining open* asks “given this is still open on day 5, what is today’s risk” rather than “does this type escalate”, which is the right structure. It does not fully solve the confounding; nothing short of a randomised trial would.

**Detention is analytical with a learned correction.** The arithmetic does the heavy lifting — overlapping trains, held or divertible, reroute penalties, caution costs, one-hop cascade. The model learns only where that arithmetic is *systematically wrong*: during peak, in known bottlenecks, when the reroute path is itself congested. This degrades safely — with no training data the correction is zero and the analytical estimate stands.

### 8.2 The cold start, handled honestly

| Stage        | Data available | Behaviour                                                                                             |
|--------------|----------------|-------------------------------------------------------------------------------------------------------|
| Rule-based   | None           | Requested duration plus a fixed buffer. Base rates from the catalogue. Fully transparent              |
| Hierarchical | Thin           | Task-type mean where samples suffice, backing off to department, then global. Blended by sample count |
| Full models  | Sufficient     | Trained models, with the rule-based version kept as a monitored fallback                              |

All three ship. The system works from day one and improves — rather than requiring a training set before it can run at all. Saying this out loud is worth more than hiding it.

### 8.3 What an officer actually sees

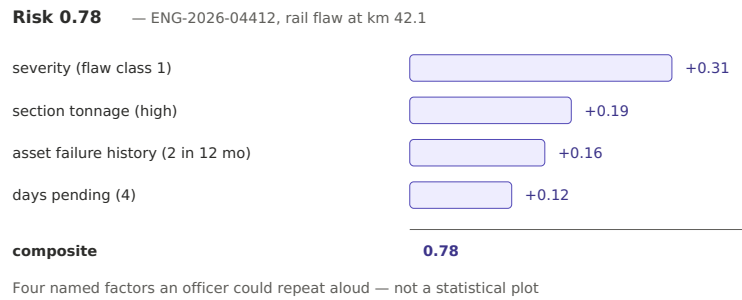


Figure 10 — A risk score decomposed into factors an officer could repeat aloud.

Not a statistical waterfall plot. Four named factors with signed contributions. If the officer cannot see why a task ranked where it did, they will not trust the ranking — and correctly so.



## 9 Planning across horizons

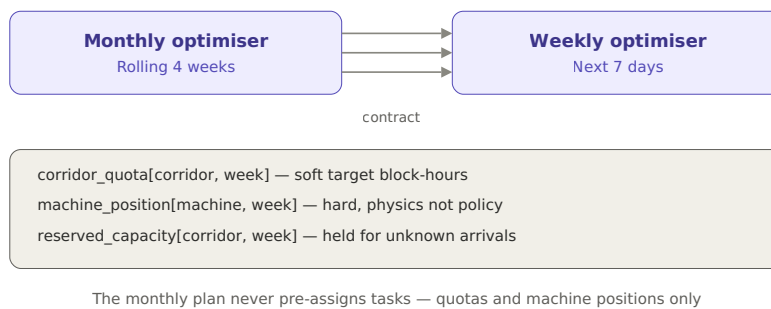


Figure 11 — The monthly plan hands down three things, and never pre-assigns tasks.

Running the weekly optimiser over thirty days does not just get slow — it produces **fiction**. Assigning a specific task to a specific Tuesday twenty-seven days out means guessing which defects will be raised in week three, where the tamper will be, and which emergencies will have eaten which windows. Precision without information is not planning.

So the monthly plan answers a different question. Not “what happens when” but “**where should capacity go**”. Its real deliverable is machine positioning: a tamper cannot teleport, so if week three carries heavy tamping demand on the Arakkonam corridor, the machine has to start moving in week two — a decision the weekly planner structurally cannot make.

**The key move.** The monthly plan does not schedule unknown defects — it **holds capacity open for them**. A corridor with a high defect arrival rate gets more block-hour quota than its known backlog justifies. Fill every hour with known work and week three's emergencies have nowhere to go.

Quotas are **soft**: if a week's real backlog exceeds its quota, the weekly planner exceeds it with a penalty, because it has better information than the monthly view did. Machine positions are **hard** — physics, not policy.

## 10 When reality breaks the plan

**An emergency is not an input to the planner. It is an input to reality.** By the time the system hears about it, the section is occupied and the crew is diverted. Nobody consulted an algorithm. The question is never “should we respond” but “given that this has happened, what is the best plan for what remains?”

An emergency consumes three things, and the one people forget is the biggest. It takes **the section**, for a few hours, and that recovers on its own. It takes **the resources** — the crew and machine diverted to attend it — and that is usually the larger disruption. A rail fracture near Ennore pulls the P.Way crew that was going to weld at Tiruvallur. The Tiruvallur section is completely fine. The work still cannot happen.



Figure 12 — What may move, and what may not.

Blocks sanctioned for tonight have gone to Control, crews are briefed, materials are at site. Re-optimising those is not clever, it is chaos. So the frozen zone is untouchable by the optimiser.

**Infeasible is not the same as suboptimal.** A frozen block whose crew was consumed by the emergency cannot happen. It is **cancelled and surfaced to the officer**, not silently rescheduled. That distinction is the difference between a system that respects operational commitments and one that treats them as suggestions.

Declaration is **manual**; re-planning is **automatic**. Only Control knows the true duration estimate and which crew was actually diverted. Auto-detection from feeds sounds impressive but risks a data glitch cancelling real work — and “the system cancelled tonight's block because of a bad feed” is exactly the failure that gets a tool switched off permanently.

And the result is presented as a **diff**: three blocks moved, one cancelled, two unchanged. That is comprehensible at three in the morning. A regenerated full schedule is not.

## 11 The scenario, end to end

### 11.1 The situation

**Week of 7-13 September. Chennai Central - Arakkonam corridor, near Tiruvallur.**

| Dept        | Work                                                              | Location          | Needs                           |
|-------------|-------------------------------------------------------------------|-------------------|---------------------------------|
| Engineering | USFD test flagged a rail flaw; welding and rail piece replacement | km 42.1, up line  | 3 h traffic block, welding crew |
| TrD         | OHE contact wire inspection; two insulator replacements           | km 40-44, up line | 1.5 h power block               |
| S&T         | Point machine servicing, overdue by 11 days                       | Tiruvallur yard   | 1 h disconnection               |

**Today:** three separate BDMS requests on three separate nights. 5.5 hours of cumulative occupation. None of the three officers knows the other two jobs exist.

### 11.2 What the system does

1. **Pools and scores.** The rail flaw scores high — serious flaw class, high-tonnage section, four days pending. The S&T job scores on overdue days. The TrD job is routine but co-located.
2. **Predicts duration.** 3 h 20 m for the Engineering job, not the 3 h requested, learned from historical welding of that class on that section.
3. **Clusters.** All three sit on TRL-AJJ. The compatibility matrix confirms OHE work may proceed inside a traffic block on a de-energised up line, and that the yard disconnection does not conflict because no heavy machine movement is planned over that turnout.
4. **Costs the windows.** Daytime  $\approx$  400 detention minutes. Wednesday 01:30-04:30  $\approx$  18 — two goods trains rerouted onto the down line, no passenger services affected.
5. **Proposes.** One shared block, Wednesday 01:30-04:30, km 40-44 up line, internally sequenced for all three departments.

### 11.3 The interruption

**Thursday 03:00 — OHE breakdown near Ennore.** The TrD emergency crew is diverted. The window and crew planned for Thursday's separate work are gone. The system flags which jobs lost their resources and regenerates the plan for the remainder of the week — presented as a diff, with the frozen zone untouched.

### 11.4 And afterwards

The rail fracture gets a temporary repair overnight, but the permanent replacement becomes a fresh high-priority task with a short hard deadline, flagged `emergency_followup`, competing for the same scarce windows as everything else.

**A failure mode to watch.** In a high-emergency period, follow-up tasks can crowd out routine work indefinitely — the backlog quietly explodes while detention figures look excellent. This is a genuine property of priority-driven scheduling, not a coding bug, and it must be monitored during evaluation.

## 12 The five points of view, illustrated

One database, one task schema, five lenses. The chain runs: **field raises and executes** → **department prioritises** → **operations sanctions** → **the board coordinates** → **management measures**. Plans flow down; actuals flow back up.

### POV 3 — The corridor board Sr.DOM · the most important screen

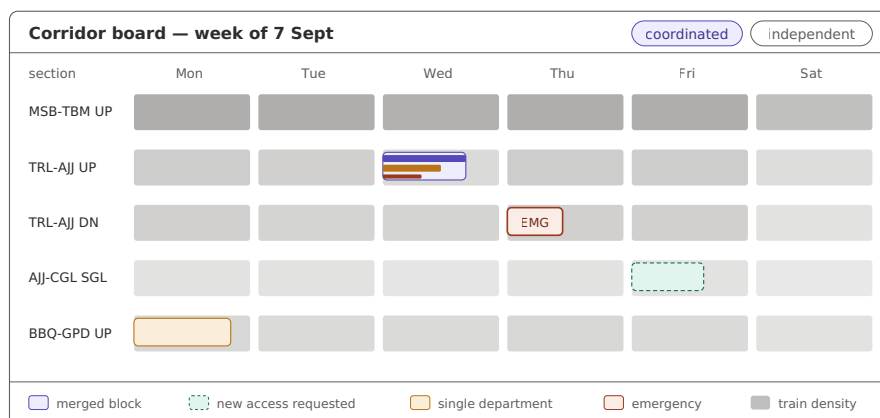


Figure 13 — Time by section, with train density as background intensity.

If only one screen were built, this is it. Train density shows as a background gradient, which makes scarce windows *obvious* rather than something to be calculated. Merged blocks show stacked department stripes; newly requested access has a distinct dashed border; emergencies overlay in red.

Clicking a block opens the alternatives the optimiser rejected, with their costs. **That is what earns trust.** “Wednesday costs 18 detention-minutes; Tuesday costs 47; daytime costs 400.” An officer who sees the comparison believes the recommendation; one who sees only the answer does not. The **before/after toggle** lives here. One switch, and the scattered bars collapse.

### POV 2 — Departmental backlog Sr.DEN · Sr.DEE/TrD · Sr.DSTE

Two panes. Left, the backlog ranked by risk, each score decomposing into named factors as in Figure 10. Right, the proposals involving their department, with approve or reject-with-structured-reason.

Rejection immediately shows what the optimiser will try instead, so rejecting feels like **negotiating rather than obstructing**. And any safety-critical task the optimiser could not fit carries a **deadline-at-risk badge** — escalation made visible rather than buried.

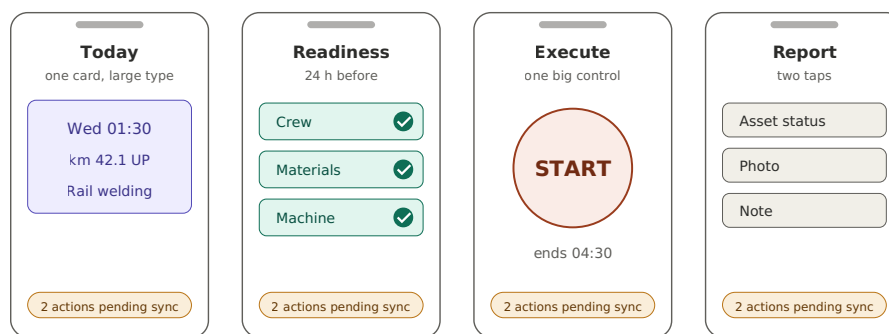
### POV 4 — The coordination board shared, all departments

One block expanded, a lane per department, the **critical path highlighted** so everyone can see whose work is setting the block duration. That is genuinely new information — today nobody knows.

Conflicts appear inline, naming the rule: “*S&T point service and ENG tamping — incompatible (CMP-018): heavy machine movement over a disconnected point machine.*” Naming the rule identifier says the system has a reason, not a preference — and gives an officer something to argue with if the rule is wrong.

Comment threads attach to the block, never floating free. This is a **block coordination board**, not a chat feature.

## POV 1 — The field application SSE / JE, all departments



Four screens. Offline-first. Tamil and English. No leaderboards, ever.

Figure 14 — Four screens, designed for 02:30 at track level.

The design brief in one sentence: usable at half past two in the morning, at track level, one-handed, with gloves on, in rain, on a five-year-old Android with one bar of signal. That eliminates most conventional interface patterns immediately.

The **readiness check** is the highest-value screen for the least code — it catches wasted blocks a full day early. The **execute** screen must never block a tap on a network call: a timer that will not start without signal teaches people to log times from memory afterwards, which destroys exactly the data the app exists to collect.

## POV 5 — The divisional dashboard DRM

Headline above everything: **“This week: 11 blocks merged into 6. 14.2 hours of line occupation saved.”** That is the system's reason for existing, stated once, large.

Below it: coordination gain over time; backlog by department with **safety breaches called out separately — that number must be zero**; the two-curve chart plotting detention caused by blocks against detention caused by speed restrictions from undone maintenance; and block utilisation, the wasted-block metric nobody currently tracks.

Plus the **persistent rejection flag**: proposals rejected three or more times, surfaced by name. That is how the system learns about constraints nobody thought to encode.

**Aggregated at section and team level only.** No individual appears anywhere in this view, and that rule is enforced in the API, not just the interface.

## 13 Ethics and responsible design

---

### 13.1 The system advises; humans sanction

No algorithm should be able to withdraw a running line from service on its own authority. Every proposal passes departmental approval and operations sanction, with reasons recorded and an audit trail retained. This is not caution — it reflects how accountability actually works on the railway, and a system that ignored it would be rejected on contact with reality.

### 13.2 Explainability is mandatory in safety systems

If the model defers a defect by two weeks, the responsible officer must see why and which factors drove it. Interpretable models are preferred over higher-accuracy opaque ones. A safety decision that cannot be explained will not be accepted, and should not be.

### 13.3 This must never become a surveillance tool

**The failure mode most likely to destroy this project.** The field application logs actual durations per crew. That data can very easily be repurposed as an instrument for disciplining workers. The moment field staff believe it is being used against them they will falsify entries — start the timer late, close it early, record convenient reasons — and the training data becomes worthless. The duration model degrades, the plans become unrealistic, and the system fails.

**Design response.** Aggregate at team and section level, never individual. Use the data for planning, never for performance review. No leaderboards, rankings or crew comparisons anywhere. Enforce it in the API. This is an uncommon case where the ethical choice and the engineering choice are *the same choice*.

### 13.4 Safety work cannot be optimised away

Safety-critical defects carry a hard deadline constraint, never a soft penalty. If it cannot be satisfied, the system escalates visibly rather than quietly rescheduling.

### 13.5 Fairness and data protection

- Night work is distributed across crews rather than repeatedly assigned to whichever team is marginally cheaper.
- Personal data collected through the field application — identity, location, working hours — is minimised, access-controlled and retained only as long as operationally necessary.
- Constraints are configurable per division. Ghat sections, saturated suburban corridors and monsoon-heavy zones have genuinely different rules; hard-coding one division's assumptions into seventy is its own kind of error.

## 14 Proving it works

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There is no ground truth. Chennai Division cannot be run for a month with the system and a month without it. So the claim comes from a **counterfactual baseline**.

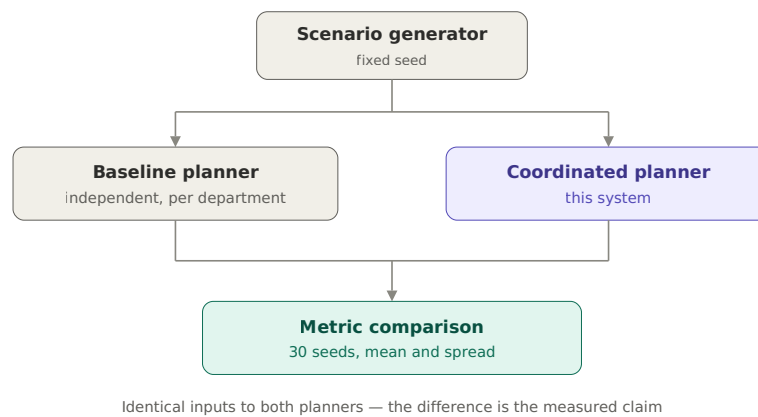


Figure 15 — Identical scenarios to both planners; the difference is the measurement.

**The baseline must be fair.** It models each department sorting its own backlog sensibly and requesting windows first-come-first-served, with no visibility of the others — plus a 10–15% incidental merge rate, because informal coordination does happen through phone calls and corridor meetings. The baseline is not random and not stupid. **The gap being measured is the cost of missing information, not the cost of incompetence** — a framing that is both true and far more persuasive.

Thirty independent seeded weeks, stratified by corridor, reporting mean and spread. Plus four ablations, each a single config flag: turn off merging, turn off the duration model, turn off availment risk, turn off detention weighting. Those prove which component actually earned its place.

| Metric tier | What is reported                                                                                                                                                    |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary     | Line occupation hours, weighted detention minutes, merge rate                                                                                                       |
| Secondary   | Tasks completed (guards against a planner that schedules nothing), backlog trend, <b>safety-deadline breaches — which must be zero</b> , days-to-attend by severity |
| Tertiary    | Block utilisation, overrun rate at the chosen buffer, re-plan recovery time                                                                                         |

**The honesty statement, said before anyone asks.** Synthetic data with parameters chosen by the team. The baseline is a model of current practice, not measured practice. The compatibility matrix and corridor block pattern are unvalidated by railway officers. **Absolute numbers are illustrative; the relative comparison is the claim.** Not “Chennai will save 14 hours” but “under identical conditions, coordinated planning dominates independent planning, and here is by how much and why.”

## 15 The demonstration narrative

Judges decide in the first ninety seconds and spend the rest confirming. So the merge image comes early, not as a climax. And throughout: **narrate the railway, not the software**. Never “now I’ll click the corridor board”; instead “Sr.DOM opens the week’s proposals.”

| Time | Beat                       | What happens                                                                                                                                                                                                                                                              |
|------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0:00 | The problem, in one screen | Corridor board set to “independent”. Three blocks, three nights. “Nobody is at fault — nobody has ever had a screen where they could see each other’s work.” Flip the toggle. The bars collapse. <b>Stop talking for two seconds.</b>                                     |
| 0:45 | Why it is hard             | Pre-empt “so you just group by location?” Show the critical path, then trigger a conflict naming rule CMP-018. Encoded domain knowledge, not generic clustering.                                                                                                          |
| 1:30 | Where the AI lives         | Decompose a risk score. Then: “Engineering requested three hours; our model predicts three twenty, learned from actual welding of this class on this section.” Then availment risk — the tamper two sections away with four hours of slack.                               |
| 2:30 | The cost model             | Alternatives panel: 18 minutes versus 47 versus 400. Then the line to practise until it is clean: <i>“Both sides of the trade-off are measured in the same unit. Taking a block costs train delay now; not doing the work costs train delay later. Minimise the sum.”</i> |
| 3:15 | The emergency              | Live, not recorded. Declare it. Show the <b>diff</b> . Then the detail that proves domain understanding: “Note what got cancelled — the Tiruvallur block. That section is fine. The crew is at Ennore. Resource loss, not section loss.”                                  |
| 4:00 | The evidence               | Thirty seeded weeks, both planners, identical inputs. Headline numbers plus <b>zero safety breaches</b> . Then the ablation in one line: “Turn merging off and the gain drops to two hours. The coordination is doing the work, not the optimiser being clever.”          |
| 4:40 | Honesty and scale          | State the limitations unprompted. Close on: “This is not a system for Indian Railways. It is the divisional planner Indian Railways would deploy seventy times.”                                                                                                          |

### 15.1 Questions to expect

| Question                                        | Answer                                                                                                                          |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| “Where’s the AI? This looks like optimisation.” | Four models named — plus prioritisation without scheduling is useless, and greedy priority never finds merges                   |
| “Railways won’t let an algorithm block a line.” | Correct. It proposes; departments approve; Sr.DOM sanctions. Full audit trail                                                   |
| “Your data isn’t real.”                         | Agreed, and stated up front. The architecture takes real feeds without redesign                                                 |
| “No history on day one.”                        | Three-stage fallback. Works from day one, improves with data                                                                    |
| “What if the officer always rejects it?”        | Persistent-rejection flag to the DRM. That is a constraint we failed to encode, and the system surfaces it                      |
| “How is this different from BDMS?”              | BDMS records requests; it does not merge them. We are a layer above it, and on conflict BDMS wins                               |
| “What about a twelve-hour possession?”          | Out of scope — planned months ahead at zonal level, and enters our system as a fixed unavailable window, not a task we schedule |

### 15.2 Failure modes to rehearse against

- **Fixed random seed, always.** Numbers that shift between rehearsal and presentation are fatal.



- **Recorded fallback for the emergency re-plan.** If the WebSocket drops, switch to video without breaking stride.
- **Do not demo the field app live.** Screenshots. Mobile demos on projectors fail more often than they work, and it is not carrying the argument.
- **If running long, cut the field app and dashboard first.** Beats 1, 5 and 6 are untouchable.

## 16 Deliberate simplifications, exclusions and future extensions

This section exists because a design is only credible if it knows its own boundaries. Everything below was a **conscious decision**, not an oversight.

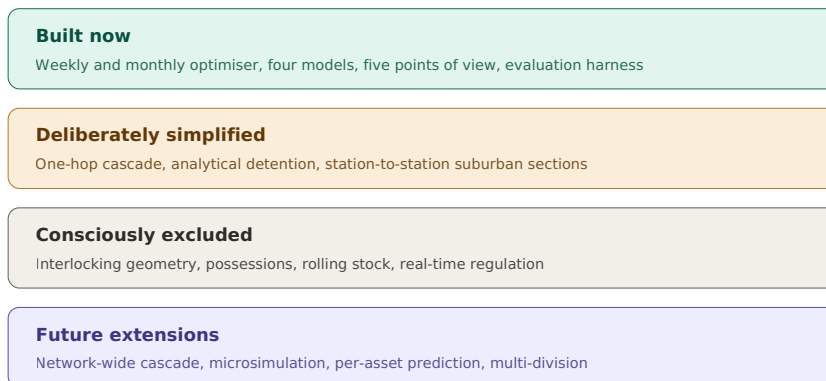


Figure 16 — Four tiers: what is built, what is simplified, what is excluded, what comes next.

### 16.1 Deliberately simplified — degraded, but present

These are modelled, but at lower fidelity than a production system would eventually want.

| What                           | How it is done now                                                                                                             | What a fuller version would do                                                                                                                                                                                                                    |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cascade delay</b>           | <b>One hop only.</b> A held train delays those immediately behind it, on the adjacent section, and there the calculation stops | Network-wide propagation — delay spreading across the division and out to neighbouring divisions, with knock-on effects on rake links and crew links. Requires a proper delay-propagation engine and materially more compute per candidate window |
| <b>Train detention</b>         | Analytical arithmetic plus a learned residual correction                                                                       | Microsimulation of train movements (in the manner of RailSys or OpenTrack), giving accurate absolute detention rather than reliable relative ranking                                                                                              |
| <b>Suburban block sections</b> | Modelled <b>station-to-station</b> , ignoring automatic signalling                                                             | True signal-spacing block sections. Realistic, but produces hundreds of sub-kilometre units that would clog the optimiser for no planning benefit                                                                                                 |
| <b>Station interiors</b>       | A <b>node resource model</b> — route count, disconnection consumes some, compatibility rules govern concurrency                | Full interlocking and route-table modelling: fouling, route conflicts, crossover positions, platform-level allocation. Needs data absent from all five source systems                                                                             |
| <b>Single-line working</b>     | A flat <b>capacity factor</b> on the surviving edge                                                                            | Explicit crossover positions determining where bidirectional working can actually begin and end, with realistic headway modelling under temporary single-line operation                                                                           |
| <b>Goods traffic</b>           | A <b>forecast by section and time band</b> , treated probabilistically                                                         | Integration with actual freight operations planning — rake availability, loading point schedules, port berth windows                                                                                                                              |
| <b>Weather</b>                 | A <b>seasonal flag</b> plus a monsoon window                                                                                   | Live meteorological feed, with short-horizon re-planning when a cyclone warning is issued — genuinely relevant on this coast                                                                                                                      |

|                             |                                                                                                    |                                                                                                                                                                                                                      |
|-----------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Escalation modelling</b> | <b>Per-defect</b> hazard, with a per-asset proactive layer as a clearly separated secondary output | Full per-asset predictive maintenance, generating inspection tasks from condition data before any defect is reported. Needs condition-monitoring instrumentation the current systems do not supply                   |
| <b>Duration uncertainty</b> | <b>Quantile regression</b> at P50, P80 and P95                                                     | Full distributional modelling with correlated uncertainty across tasks in a merged block — because when one job overruns, the others in the same block often do too                                                  |
| <b>Optimiser paradigm</b>   | <b>Window filling</b> plus a local refinement pass                                                 | Free-form block construction with interval variables and no-overlap constraints against train paths, able to invent windows nobody pre-computed. Strictly more powerful, considerably harder to debug and to explain |

## 16.2 Consciously excluded — correctly outside the boundary

| Excluded                                   | Why this is the right call                                                                                                                                                                                       |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Rolling stock maintenance</b>           | Coaches, periodic overhauls, sick lines and shed scheduling belong to Mechanical / C&W, run on different systems, and need no block. A coach comes to the shed; the shed does not stop trains                    |
| <b>Possessions and mega-blocks</b>         | Long planned occupations with the timetable formally modified around them. Planned months ahead at zonal level. They enter this system as <b>fixed unavailable windows</b> , which is the correct interface      |
| <b>Real-time train regulation</b>          | Minute-by-minute decisions about which train goes first are Control's on-the-day function. A weekly planner deciding whether a block happens Wednesday or Thursday operates at a completely different resolution |
| <b>Financial costing</b>                   | Everything is denominated in weighted delay minutes rather than rupees. Delay minutes are what railway operating staff already reason in, and avoid a whole layer of contestable monetary assumptions            |
| <b>Materials procurement</b>               | A readiness flag is consumed as an input; the supply chain behind it is not modelled. Integrating with materials management is a separate system boundary                                                        |
| <b>Cross-division through traffic</b>      | Trains crossing divisional boundaries have effects the divisional planner does not see. The zonal layer handles machine allocation only. Full multi-division coordination is a genuine extension, not a gap      |
| <b>Level crossing and tower wagon work</b> | Real activities, but they rarely drive the block-planning decision and would add task types without changing the optimisation shape                                                                              |

## 16.3 Future extensions — in rough order of value

| # | Extension                              | What it would add                                                                                                                                                                                                                                  |
|---|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Real data integration</b>           | Replace the synthetic generator with live adapters to TMS, SMMS, TDMS, COA and BDMS. Everything else is designed to accept this without redesign. Also replaces the modelled corridor block pattern with the real working timetable table          |
| 2 | <b>Compatibility matrix validation</b> | Sign-off from P.Way, TrD and S&T officers on every rule. The <code>validated_by</code> column exists and is deliberately blank. This is the single highest-value non-technical step                                                                |
| 3 | <b>Multi-division rollout</b>          | The architecture is already division-agnostic — <code>division_id</code> is carried throughout. Shared models with a division feature would shorten cold start dramatically for division seventy-one, using Chennai's history as transfer learning |

|    |                                            |                                                                                                                                                                                                                                                |
|----|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4  | <b>Network-wide cascade modelling</b>      | The largest fidelity gain available. Replaces one-hop propagation with a proper delay-propagation engine, materially improving detention accuracy on dense corridors                                                                           |
| 5  | <b>Per-asset predictive maintenance</b>    | Promote the proactive layer from secondary suggestion to a first-class task generator, driven by condition monitoring rather than reported defects. Shifts the system from scheduling known work to anticipating unknown work                  |
| 6  | <b>Live weather integration</b>            | Cyclone and heavy-rain warnings triggering short-horizon re-planning. On this coast, between October and December, this is not a marginal feature                                                                                              |
| 7  | <b>Crossover-aware single-line working</b> | Model where bidirectional working can physically start and end, replacing the flat capacity factor. Improves the accuracy of double-line block costing considerably                                                                            |
| 8  | <b>Free-form block construction</b>        | Move from window filling to interval-variable scheduling, allowing the optimiser to find windows nobody enumerated. Better solutions, at the cost of explainability — worth doing only once the current system is trusted                      |
| 9  | <b>Materials and machine logistics</b>     | Extend from a readiness flag to actual supply-chain integration, so the optimiser can schedule around a rail delivery rather than waiting to be told it has not arrived                                                                        |
| 10 | <b>Zonal and national aggregation</b>      | Roll divisional plans into a zonal view for machine fleet allocation and campaign planning across division boundaries                                                                                                                          |
| 11 | <b>Learning the compatibility matrix</b>   | Use the persistent-rejection signal to <i>propose</i> new compatibility rules from observed officer behaviour — the system inferring constraints nobody wrote down. Must remain suggestion-only; safety rules should never be silently learned |
| 12 | <b>Possession planning module</b>          | Extend upward into long-horizon possession planning, currently treated only as fixed unavailable windows                                                                                                                                       |

**The pattern in this list.** Every simplification was chosen so that removing it later is an *extension*, not a rewrite. One-hop cascade becomes n-hop by replacing one function. Analytical detention becomes simulation by swapping the estimator behind the same interface. Window filling becomes block construction by changing the CP-SAT variable definitions while the clustering and scoring layers stay untouched. That is the property worth defending in a viva.

## 17 Honest limitations

| Limitation                                      | Position                                                                                                                                                                                                                                           |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>No access to real operational data</b>       | Synthetic generation grounded in real Chennai geography, stated openly. The architecture is designed so real feeds substitute without redesign                                                                                                     |
| <b>Compatibility rules are approximations</b>   | Authored from published safety principles and general practice, not divisional standing orders. Flagged as requiring officer validation before deployment                                                                                          |
| <b>Corridor block pattern is modelled</b>       | Generated from a documented rule set held in configuration, not from the real working timetable. Replaced wholesale on deployment                                                                                                                  |
| <b>Duration model has no history initially</b>  | Cold start is real. Three-stage fallback handles it, and this is stated rather than hidden                                                                                                                                                         |
| <b>Detention estimation is simplified</b>       | Section-level analytical estimate with learned correction. Adequate for ranking options, not for absolute prediction                                                                                                                               |
| <b>Escalation modelling carries confounding</b> | Dangerous defects get fixed fastest, biasing the observed relationship. Hazard-conditional-on-open mitigates but does not eliminate this                                                                                                           |
| <b>Data integration is the real difficulty</b>  | Not computation. Three departments, heterogeneous schemas, uneven data quality across seventy divisions. The unified task schema is the mitigation, and it is deliberately the first deliverable                                                   |
| <b>Adoption risk</b>                            | A technically correct system that is organisationally rejected is worth nothing. Hence the approval workflow, structured rejection reasons, explainable scores and surveillance safeguards — these matter more than the optimiser's optimality gap |

## 18 Glossary

| Term                     | Meaning                                                                                                         |
|--------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Availment</b>         | Whether a sanctioned block was actually used. A block granted but not used is a wasted block                    |
| <b>BDMS</b>              | Block & Disconnection Management System — where block requests are formally filed. Remains the system of record |
| <b>Block section</b>     | The stretch of one line between two consecutive block stations. The unit over which a block is granted          |
| <b>COA</b>               | Control Office Application — timetable, traffic and corridor availability                                       |
| <b>Corridor block</b>    | A maintenance window pre-planned into the working timetable                                                     |
| <b>Detention</b>         | Train delay, measured in minutes. The unit of this system's objective function                                  |
| <b>Disconnection</b>     | Temporary disconnection of signalling gear from the interlocking, with Station Master authority                 |
| <b>OHE</b>               | Overhead Equipment — the electrical distribution above the track                                                |
| <b>P.Way</b>             | Permanent Way — the track itself, and the Engineering department that maintains it                              |
| <b>Possession</b>        | A long planned occupation with the timetable modified around it. Distinct from a block, and out of scope here   |
| <b>Power block</b>       | De-energisation of the OHE. Implies a traffic block on an electrified line                                      |
| <b>Speed restriction</b> | A caution order limiting speed over a defective stretch, imposed until repair. Costs delay every day it stands  |
| <b>SMMS / TDMS / TMS</b> | The maintenance systems of S&T, Traction Distribution and Engineering respectively                              |
| <b>Traffic block</b>     | Withdrawal of one line from train service for a defined period                                                  |
| <b>USFD</b>              | Ultrasonic Flaw Detection — how internal rail defects are found                                                 |

**In closing.** The current process is decentralised, manual, and was designed for a network with slack. The slack is gone. This system does not replace the people who plan blocks or the systems they already use — it gives them a shared, scored, costed view of work that is presently invisible across departmental boundaries, and proposes plans they remain entirely free to reject. The measurable outcome is hours of corridor time returned to train operations, with no maintenance task sacrificed and no safety deadline missed.